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THE NEXT REVISION SHOULD IMPLEMENT A SYSTEM BEHAVIOR IF THE HITL BOARD LOSES POWER

A WATCHDOG FOR MUX FAILURES WOULD BE IMPLEMENTED IN NEXT REVISION

GENERIC RES, CAP, AND OTHER COMPONENT TYPES USE GENERIC VALUES FOR NOW UNTIL BOM IS FILLED OUT

OSCILLATOR LOAD CAPS ARE PROTOTYPE VALUES; QUALIFY STARTUP, FREQUENCY

REVISIONS

ZONE	LTR	DESCRIPTION	DATE	APPR.
	-	RELEASED PER ENGR ECO - XXXXXXXX	XXXXXXXX	

TABLE A

REFERENCE DESIGNATION	POWER INPUT TO PINS							

HITL - APPLICATION STARTUP / SOFTWARE NOTESBasis: HITL(4).pdf SH2-SH7 and supplied native sheets; 09 SEP 2026.These are new implementation notes, not a review of the cover sheet.1. TARGET / CLOCKSSTM32H743ZIT6, LQFP144. HSE = 25 MHz; LSE = 32.768 kHz.Start diagnostics from reset HSI; add a validated clock tree for the exactsilicon revision, voltage scale and flash latency. All clock waits needbounded timeouts. W5500 has its own 25 MHz crystal.2. INITIAL OUTPUT STATEPreload GPIO levels before making them outputs: Ethernet reset PG1 = LOW;CAN standby PG10 = HIGH; onboard CS pins PF6, PE4 and PB12 = HIGH;buzzer PB0 = LOW; indicator PC0 = LOW. Synthetic DSHOT/UART/MAG branchesmust not drive peers until their mode has been explicitly authorized.3. IO EXPANDER FIRSTU10 TCA9535 is on I2C2: PB10=SCL, PB11=SDA, PG6=active-low INT.Use 100 kHz initially. Address is 7-bit 0x20 (0x40 only for APIs requiringa left-shifted address). Write each register explicitly, in this order: 0x02 OUTPUT0 = 0x00 0x03 OUTPUT1 = 0x38 0x04 POLARITY0 = 0x00 0x05 POLARITY1 = 0x00 0x06 CONFIG0 = 0x40 0x07 CONFIG1 = 0xC0Read back all six registers, then both input registers. This selects realpaths on powered muxes and sets the common-anode RGB channels off. P06is an input until the external PDB-GPIO direction is defined. P06 isHITL2PDB_GPIO; it is NOT an STM32 GPIO in this revision.4. MUX BIT MAPPort 0 bits 0..7: GPS_RX_SEL, GPS_TX_SEL, RF_RX_SEL, RF_TX_SEL,MAG_SCL_SEL, MAG_SDA_SEL, HITL2PDB_GPIO, PDB_DISCONNECT.Port 1 bits 0..7: PDB_SEL, ESC_SEL, ESC_DISCONNECT, RGB_R, RGB_G,RGB_B, unused input, unused input. SELECT=0 real; SELECT=1 HITL.DISCONNECT=1 isolates TMUX1574 paths. RGB bits are active low.Use one register owner and masked writes; change both MAG select bitsin one port write only while SCL/SDA are idle. R95 defines P06's idle bias.5. PERIPHERAL BRING-UPEthernet: SPI5 PF6/7/8/9 = CS/SCK/MISO/MOSI; PG0=INTn, PG1=RSTn.After valid rail/clock conditions, pulse reset LOW for 2 ms, release,wait at least 2 ms, then read VERSIONR=0x04. Start SPI mode 0 at <=1 MHz.FRAM: SPI4 PE4/PE2/PE5/PE6 = CS/SCK/MISO/MOSI; 64 KiB, mode 0.SD: SPI2 PB12/13/14/15 = CS/SCK/MISO/MOSI; PG7=card detect.Initialize SD at <=400 kHz; this socket is SPI, not SDMMC.CAN: FDCAN1 PD0=RX, PD1=TX. Drive PG10 LOW only after configuration.Rates, network addresses and peer protocols must come from board config.6. MODE OWNERSHIPOne mux manager owns all selector writes. Boot into unarmed diagnostics;never restore an armed/spoofing state automatically from nonvolatile memory.Prepare the selected waveform/protocol before enabling a HITL path. Theunselected MCU branch cannot monitor real traffic through a disconnectedswitch. MAG target address/register behavior requires the actual sensor spec.Apply conditional MAG rewire 08 only after confirming P4=FMU, P6=GPS/MAG.7. MEMORY / DMAPlace DMA buffers in SRAM accessible to the selected DMA controller,not blindly in DTCM. SPI6/BDMA needs a D3-accessible buffer. Handle theCortex-M7 32-byte cache lines using a deliberate cache-maintenance or MPUpolicy. SD/network logging must not block timing-critical signal generation.8. RESET / PROGRAMMING / FAULT LIMITSMCU NRST does not reset the TCA9535. A warm restart can inherit old muxselections. Disconnect actuator/external signal harnesses and cold-cyclethe complete board before ROM USB DFU. ROM peripheral probing may touchpins also assigned to DSHOT. Verify BOOT0/reset entry on the actual chip.Hardware change 09 removes PA9 VBUS detection: HAS_VBUS_GPIO_SENSE=false.USB is for programming; measure attach/DFU current with accessories absent.There is no unpowered bypass or hardware watchdog override in this revision.At mux power loss all signal paths isolate. On a control-bus fault, quiesceMCU-generated outputs and report mode UNKNOWN; do not claim software canforce real pass-through when the expander is unreachable.

HITL firmware handoff for Codex

CONTRACT NO	
DWN BY	
CHECK BY	JON TOLDEN
APPV BY	
PROJECT SUPERVISOR	DR. HEDAYATIPOUR
MKS Rev MKS Date	

DEESCPROTO - LONG BEACH, CA
ELECTRICAL ENGINEERING

TITLE:

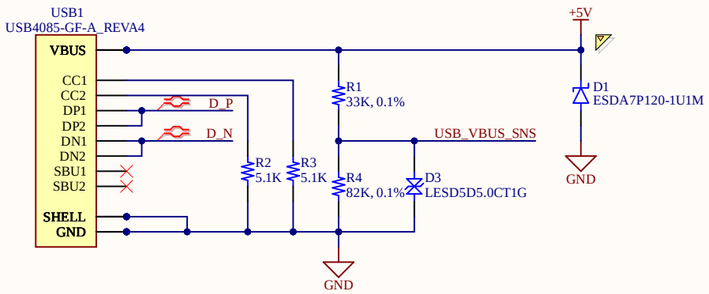
HUMAN IN THE LOOP FOR UAS MULTIROTOR PLATFORM

SIZE		DWG NO	REV
B			-
SCALE:	VIEWDRAW	SHEET 1 OF 7	

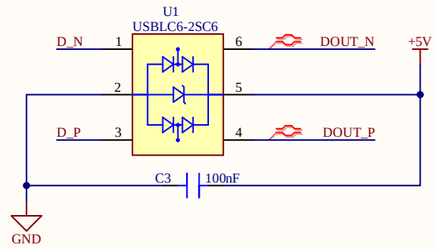
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

POWER SCHEME

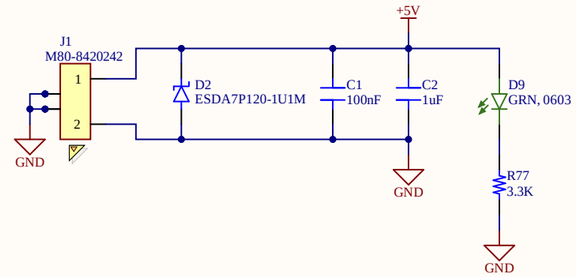
USB-C



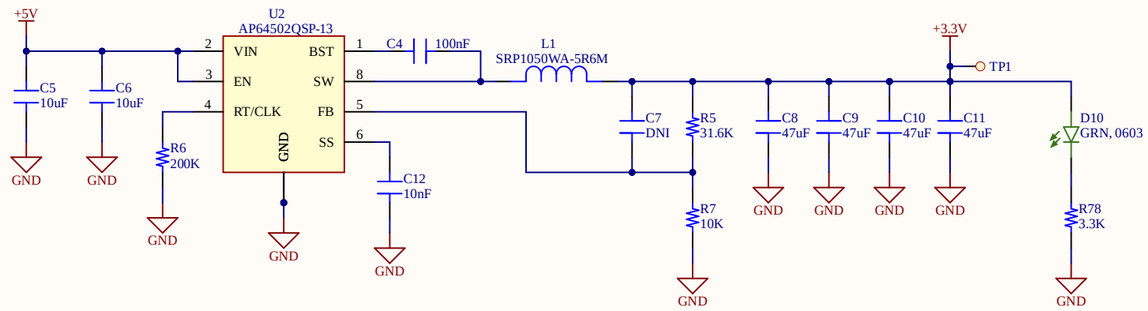
USB ESD



+5V FROM PDB



SWITCHING REGULATOR

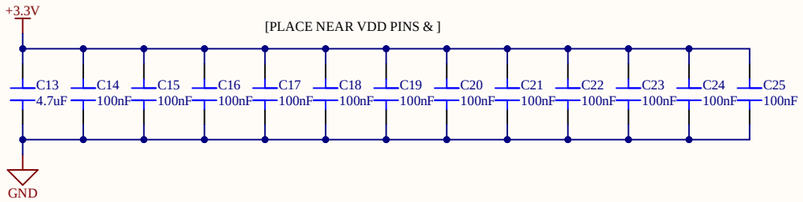


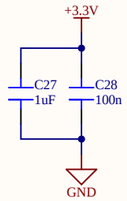
SIZE B	DWG NO		REV -
SCALE:	VIEWDRAW	SHEET 2 OF 7	

UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

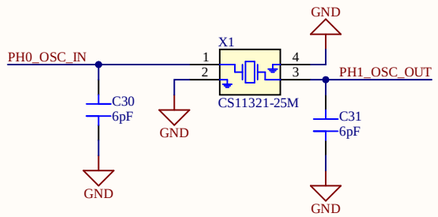
STM32H743ZIT6

DECOUPLING STM32H7

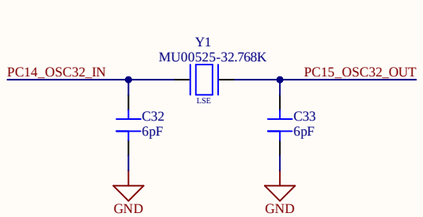




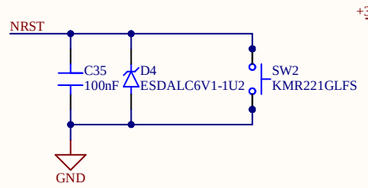
HSE

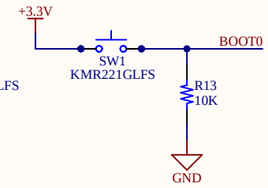


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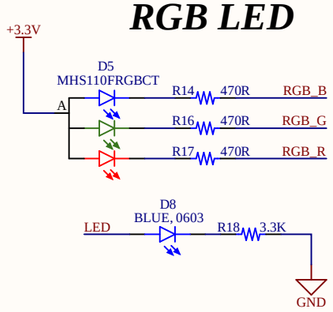


NRST & BOOT

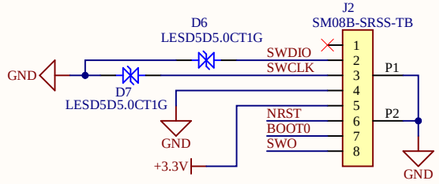


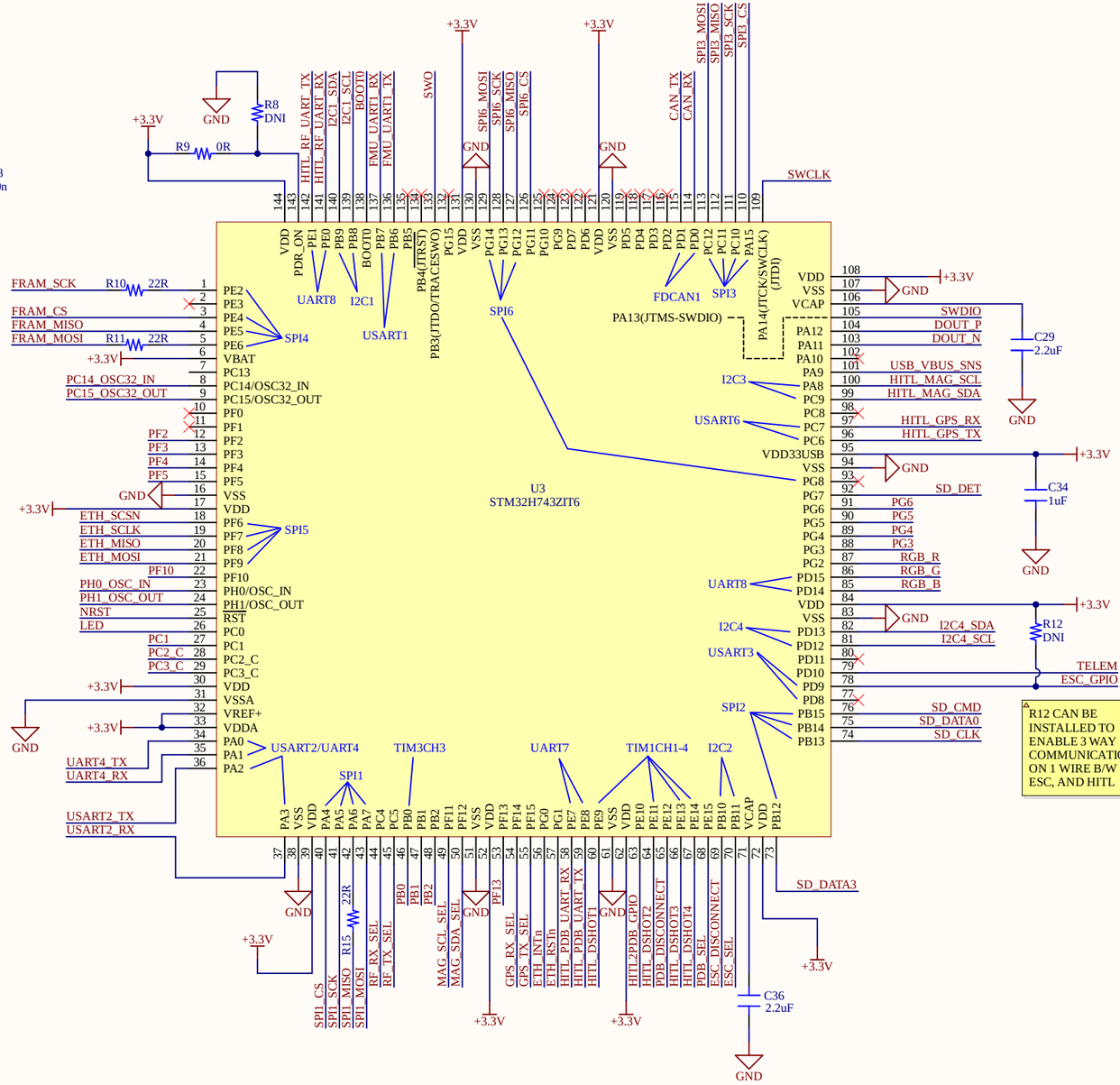


RGB LED



PRGM CONN





	SIZE B	DWG NO	REV -
SCALE:	VIEWDRAW	SHEET 3 OF 7	

LOGIC LOW == SYS NORMAL FUNCTION

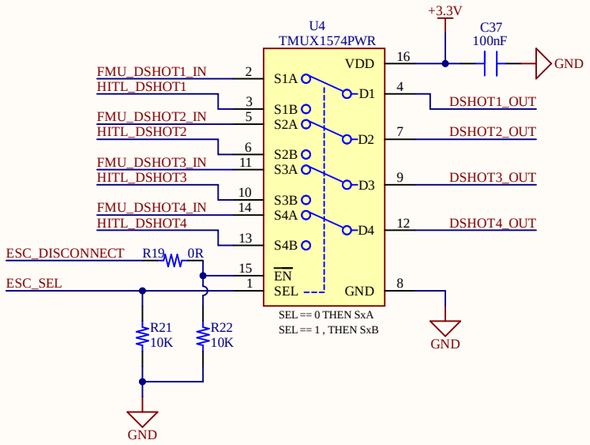
LOGIC HIGH == HITL SPOOFED SIGNAL

UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

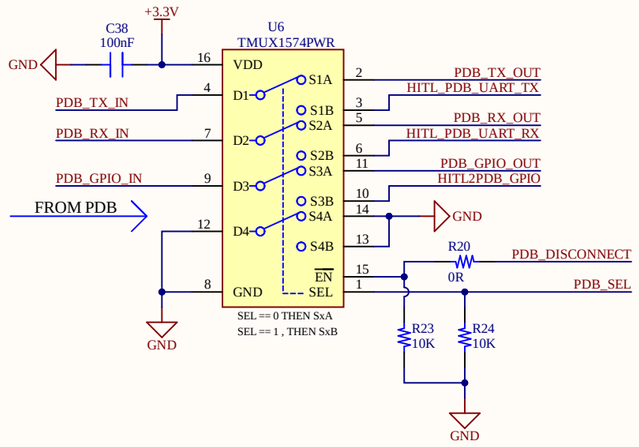
SIGNAL SPOOFING

WITH VALID MUX SUPPLY AND SEL/EN AT THEIR DEFAULT LOW
A WATCHDOG FOR MUX FAILURES WOULD BE IMPLEMENTED IN NEXT REVISION

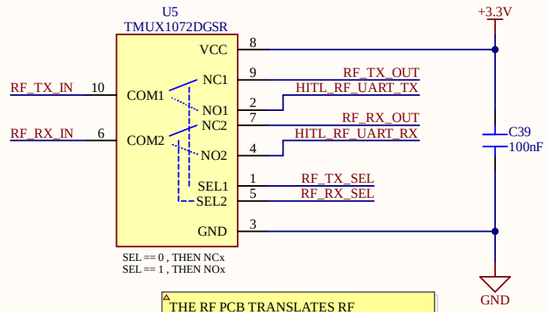
DSHOT MUX



PDB MUX



RF MUX

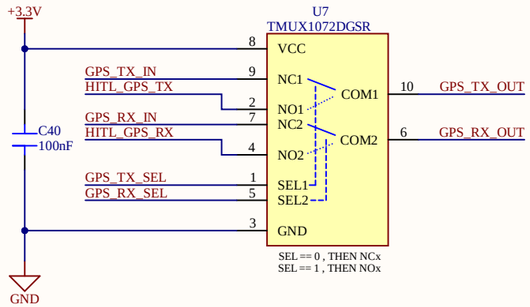


THE RF PCB TRANSLATES RF COMMUNICATION TO UART THROUGH INPUT CONNECTOR J3.

THE HITL CAN SELECT IF ITS OWN SPOOFED UART OR THE RAW ORIGINAL UART FROM RF PCB GOES THROUGH MUX.

THIS UART SIGNAL LEAVES THROUGH CONNECTOR J5 TO THE FMU

GPS MUX

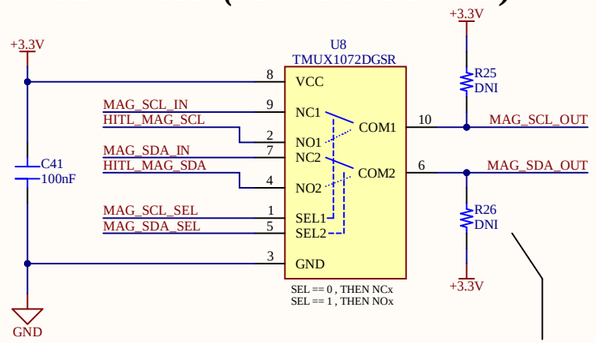


RAW GPS AND MAG DATA COMES IN THROUGH CONNECTOR P3 FROM GPS BOARD.

HITL HAS THE OPTION TO GENERATE A SPOOFED VERSION OF THIS DATA ON 2 OF ITS OWN UART PINS.

TO SEND THAT DATA AND BLOCK GPS PCB, THE HITL CONTROLS U9 TO DECIDE IF THE RAW SIGNAL OR HITL'S SPOOFED SIGNAL IS SELECTED BY U9 MUX.

MAG MUX (FROM GPS PWBA)



IF MAG I2C LINES ON GPS BOARD DO NOT HAVE PULL RESISTORS INSTALL 4.7K TO R47 AND R49

RAW GPS AND MAG DATA COMES IN THROUGH CONNECTOR P3 FROM GPS BOARD.

HITL HAS THE OPTION TO GENERATE A SPOOFED VERSION OF THIS DATA ON 2 OF ITS OWN I2C.

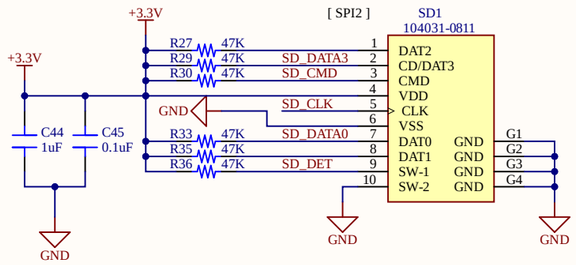
TO SEND THAT DATA AND BLOCK GPS/MAG PCB, THE HITL CONTROLS U9 TO DECIDE IF THE RAW SIGNAL OR HITL'S SPOOFED SIGNAL IS SELECTED BY U9 MUX.

SIZE	DWG NO	REV
B		-
SCALE:	VIEWDRAW	SHEET 4 OF 7

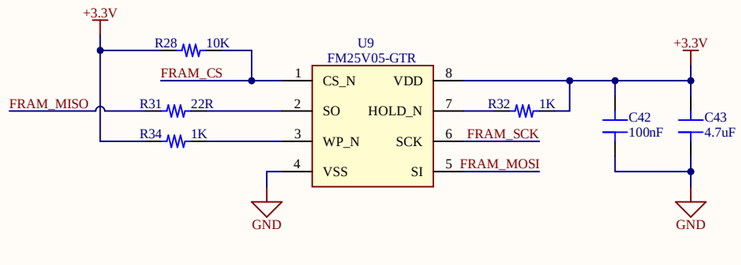
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

MEMORY

SD CARD



FRAM

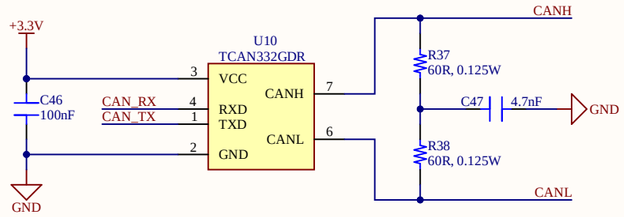


SIZE B	DWG NO		REV -
SCALE:	VIEWDRAW	SHEET 5 OF 7	

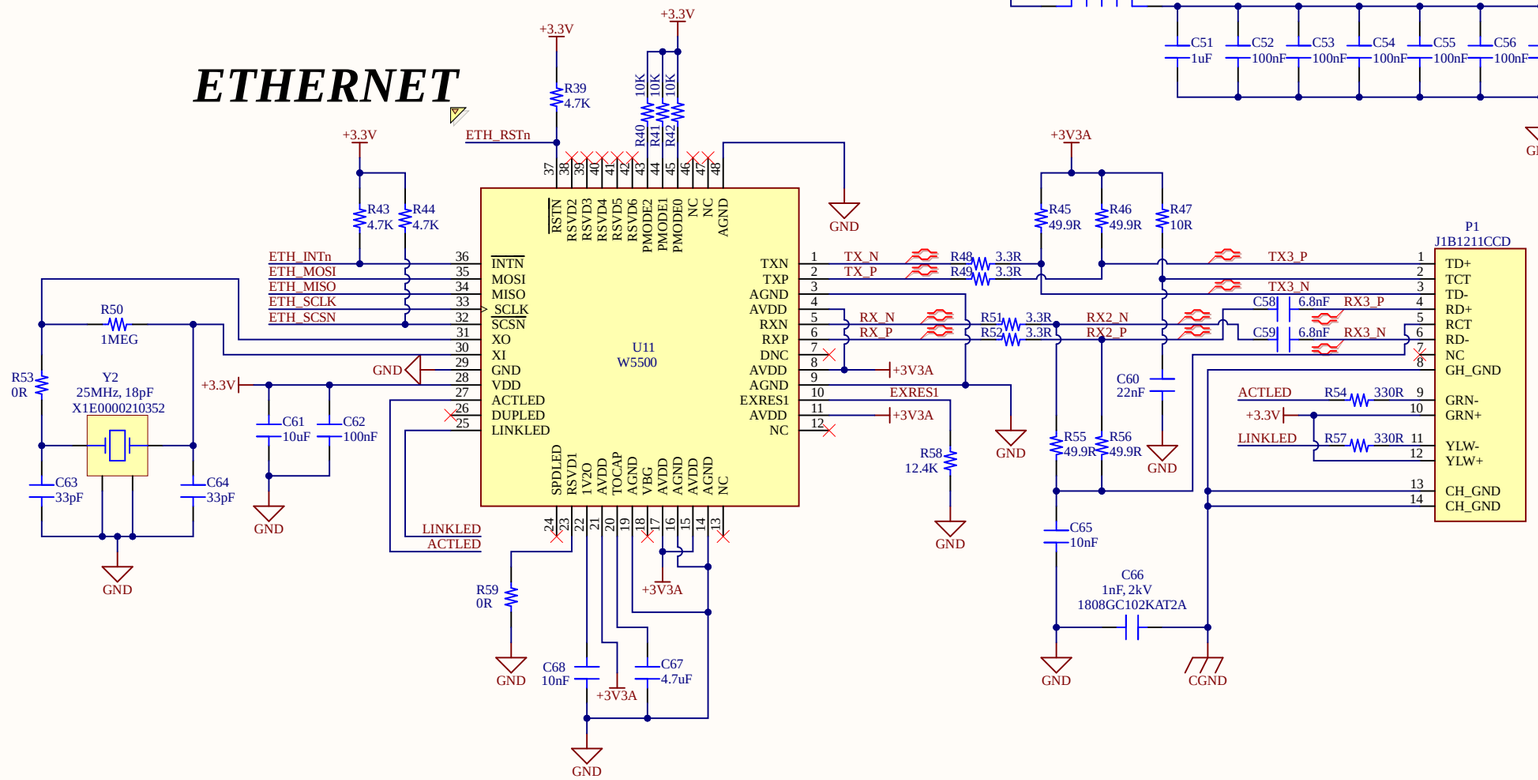
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

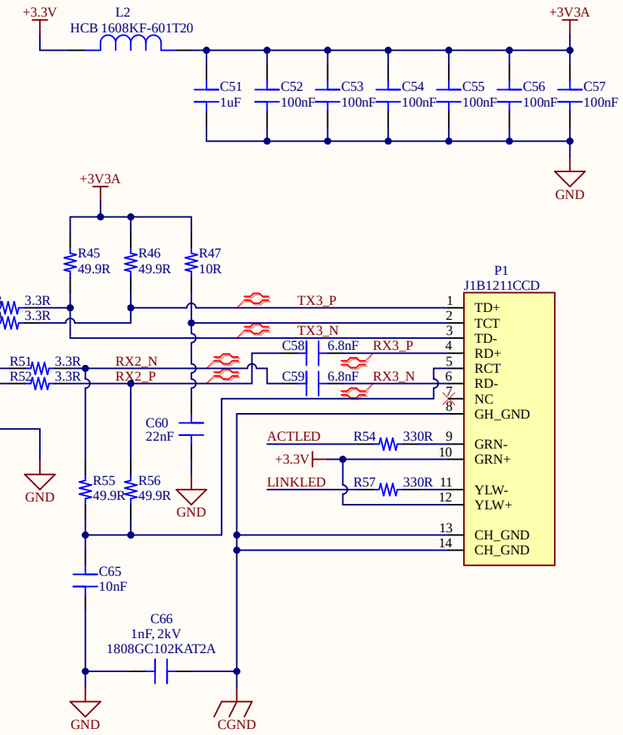
INTERFACE PT1 - CONNECTIVITY

CAN BUS



ETHERNET



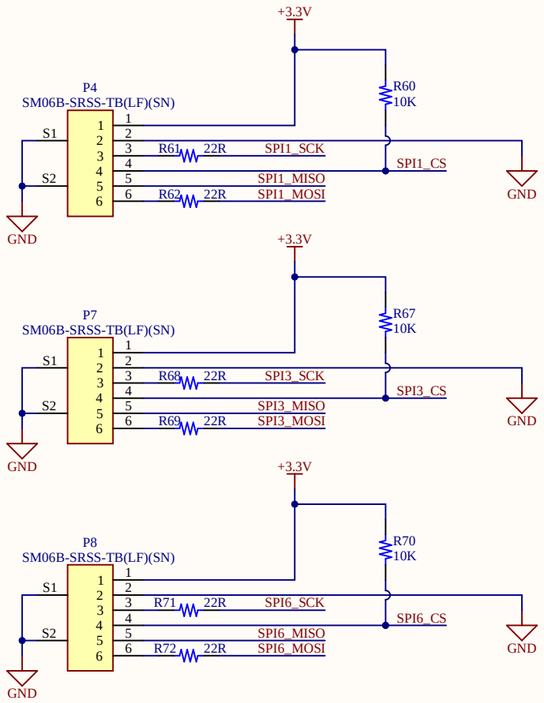


SIZE B	DWG NO		REV -
SCALE:	VIEWDRAW	SHEET 6 OF 7	

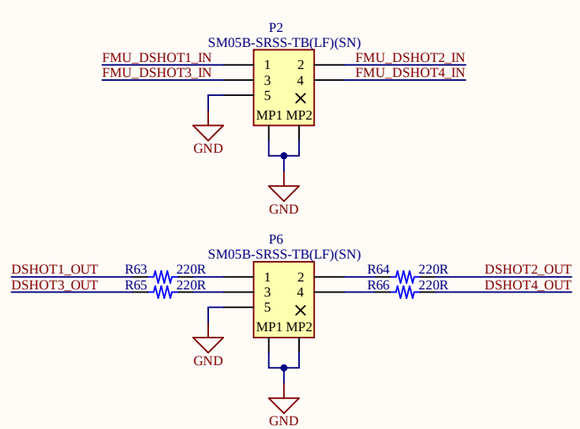
UNLESS OTHERWISE NOTED:
ALL RESISTORS ARE 1%, 0.1W
ALL CAPACITORS ARE 10%, 50V

INTERFACE PT2 - CONNECTORS

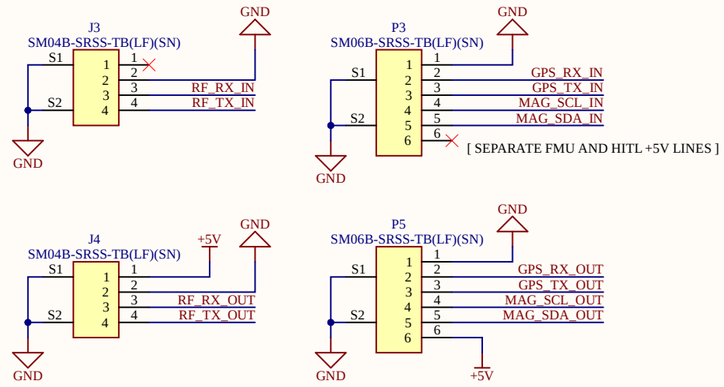
SPI CONNECTORS



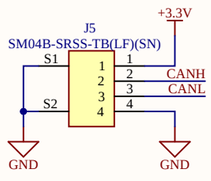
DSHOT CONNECTORS



RF CONTROLLER / GPS



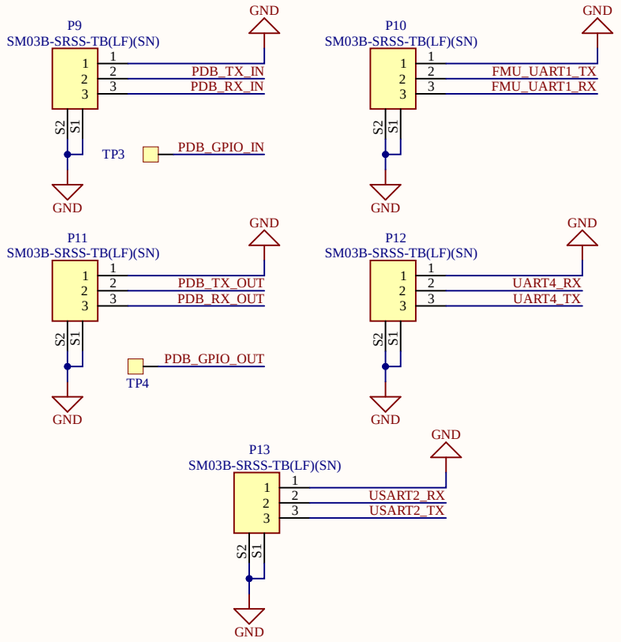
CAN BUS



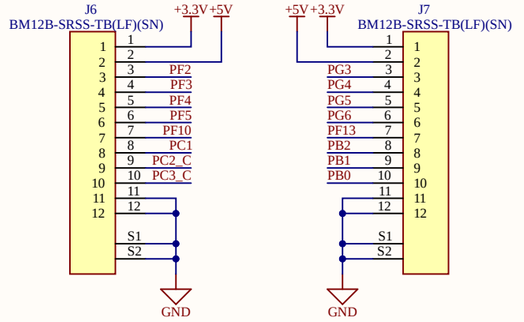
UART CONNECTORS

UART7: PDB TELEMETRY

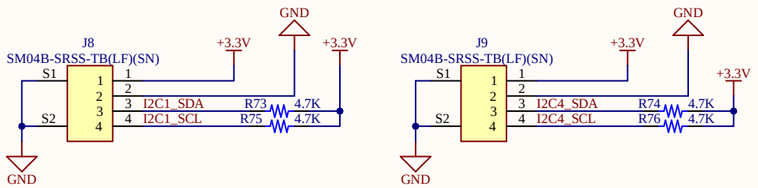
UART 1: FMU TELEMETRY



EXTRA GPIO PINS



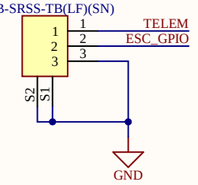
I2C CONNECTORS



CAREFUL SHARING +3.3V BETWEEN 2 +3.3V SOURCES ON 2 DIFFERENT BOARDS.

COMMS WITH ESC

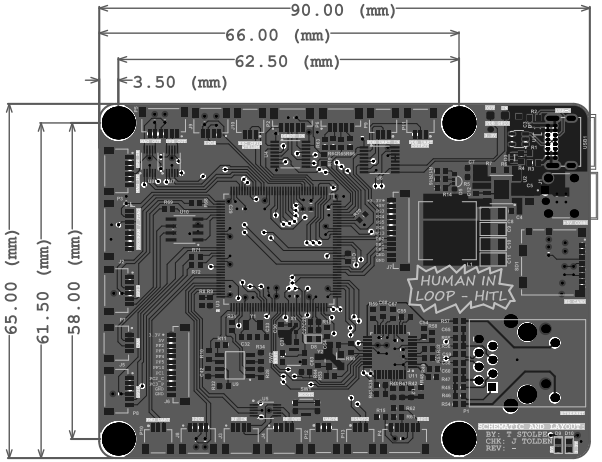
THIS ENABLES HITL TO HAVE DIRECT COMMUNICATION OPTIONS WITH ESC.



EXAMPLE: If the Mux's give us some kind of issue, we can do human in the loop "hard-coded." So for the ESC, the HITL would (at the same) time pull the shared GPIO with the ESC from low to high and tell the FMU to stop DSHOT over their shared UART lines. Then the HITL can use the telem pin on the ESC to send motor instructions while also using the shared GPIO line for 2 way direct comms with ESC.

SIZE	DWG NO	REV
B		-
SCALE:	VIEWDRAW	SHEET 7 OF 7

LAYER 1: SIGNAL
LAYER 2: GND
LAYER 3: SIGNAL
LAYER 4: POWER
LAYER 5: GND
LAYER 6: SIGNAL



DESIGN INFORMATION			
MIN. TRACK WIDTH:	____ MIL		
MIN. CLEARANCE:	____ MIL		
MIN. VIA PAD SIZE:	____ MIL		
MINIMUM ANNULAR RING	0.05mm (2MIL)	EXTERNAL	
PER	IPC-D-275 CLASS 2 LEVEL C		
REGISTRATION TOLERANCES:	METAL +/- ____ MIL, HOLES	+/- 3 MIL	
HOLE SIZE TOLERANCE (UNLESS OTHERWISE SPECIFIED):		+/- 3 MIL	
MATERIAL:			
[ ] FR-408	[X] FR-4 High Tg	[ ] OTHER ____	
THICKNESS:	[X] 62 MIL (1.2mm) +/-10%	[ ] OTHER ____	
TOLERANCE:	[X] ANSI IPC-4012 TYPE 3 CLASS 2		
	[ ] OTHER +/- ____		
BOW & TWIST:	[X] ANSI IPC-4012 TYPE 3 CLASS 2		
	[ ] OTHER +/- ____		
DRILLING:			
REFERENCE:	[X] AS SHOWN	[X] NC_DRILL FILES	
PTH COPPER THICKNESS:	[X] 25-38 um	[ ] OTHER ____	
BOARD FINISH:			
SILKSCREEN:	[X] TOP	[X] BOTTOM	
SILKSCREEN COLOR:	[X] WHITE	[ ] OTHER ____	
SOLDER RESIST COLOR:	[ ] GREEN	[X] OTHER RED	
	[X] MATTE	[ ] SEMI-GLOSS	
SURFACE FINISH:	[X] IMMERSION GOLD (ENIG)	[ ] ENEPIG	
[ ] IMM. TIN/SILVER OR EQUIV		[ ] OTHER ____	
ARRAY/PANEL:	[ ] CUT AND TRIM PER MY BOARD OUTLINE		
	[ ] N.C. ROUTE	[X] V- SCORE	
CERTIFICATION:	MATERIALS AND WORKMANSHIP FOR ALL PCBS TO MEET OR EXCEED THE REQUIREMENTS OF:		
[X] ANSI/IPC-A-600F CLASS -2-		[ ] 1 [X] 2 [ ] 3	
[X] RoHS		[ ] OTHER ____ PER ORDER	
ALL BOARDS MUST MEET OR EXCEED EIA-RKO REQUIREMENTS.			
ADDITIONAL REQUIREMENTS:			
MICROSECTION:	[ ] YES		
BARE BOARD ELEC. TEST:	[ ] NONE	[X] REQUIRED	[ ] PER ORDER

DEESC

PROJECT TITLE:

DESIGNED FOR:

FILE NAME: HITL R DRAFT 3 PCB.PcbDoc

1000.00mil

COMPONENTS MARKED 'DNI' SHOULD NOT BE POPULATED.

ASSEMBLY VARIANT: [No Variations]

PCB VIEWED FROM TOP SIDE		
SOFTWARE BY: BAO PHAM		
POWER SCHEME BY: AUTUMN AGUILAR	GENERATED : 10/14/2026 9:26:04 PM	

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PROJECT MANAGER: JON TOLDEN	L-YOUT BY:
SCALE: 0.72	ALTIUM DESIGNER VERSION: 26.10.1.5